Mary Kate Evangelidou

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SUMMARY

Molecular biologist with a BSc in Molecular Biology & Genetics, a year of thesis research at BRFAA on gene regulatory networks in breast cancer cells, and two years of training in DNA cloning. Long drawn to molecular neuroscience, developmental and behavioural biology, and keen to contribute to research in these fields. Also trained as a software engineer at 42 Heilbronn (C, C++, Python, TypeScript), bringing programming skills that can support data analysis and the building of research tools.

EDUCATION

Mar 2023 – Sep 2026

Software Engineering · 42 Heilbronn ↗

Germany · peer-to-peer and project-based curriculum · full scholarship

- 25 projects: C and Unix systems (processes, threads, memory, a bash-like shell), C++, an HTTP server, Docker
- PongJam ↗, graduation project: realtime multiplayer web platform, team of 5; built the frontend (TypeScript, Tailwind CSS) and designed the UI/UX
- miniRT ↗: ray tracer in C, team of 2; built the scene-file parser and the switch between display modes; wrote the installation and user documentation

Sep 2018 – Jul 2023

BSc Molecular Biology & Genetics ↗

Democritus University of Thrace (DUTH), Greece · 240 ECTS

- Final grade 9.22/10 ("Excellent", top band 8.5–10), ranked 3rd in my year
- Thesis (10/10): *Study of Gene Regulatory Networks Affecting Metabolism and DNA Repair in Cancer Cells* ↗
- Selected courses: Molecular Neurobiology, Developmental Biology, Genomics, Systems Biology, Bioinformatics, Biostatistics
- Certificate of Pedagogical and Teaching Competence: 8 courses, incl. microteaching and adult education
- Selected presentations and slide design ↗: SNAREs in synaptic transmission, epigenetics in forensics, Nobel Prize in Chemistry 2015, epidemic modelling

RESEARCH EXPERIENCE

Mar 2022 – Feb 2023

Undergraduate researcher · BRFAA

Biomedical Research Foundation of the Academy of Athens, Greece

- Investigated whether the transcription factor Prox1 regulates DNA-repair genes in MCF7 and MDA-MB-231 breast cancer cells
- Selected 13 genes across DNA-repair pathways from the lab's RNA-seq data on mouse Neuro2A cells; designed qPCR primers for 15 genes in total
- Mammalian cell culture, adenoviral overexpression (GFP control), RNA extraction, cDNA synthesis, qPCR (ΔΔCq, 3 biological replicates)
- 8 of 15 genes changed significantly in each line, adding detail to the possible dual role of Prox1 in breast cancer

Feb 2020 – Dec 2021

Laboratory research trainee · DUTH

Democritus University of Thrace, Greece

- Cloned partner genes of an essential developmental protein into multiple vectors with different fluorescent tags
- PCR, restriction digests, gel electrophoresis, electroelution, ligation, bacterial transformation and culture
- Documented every experiment in a 160-page handwritten lab notebook

SKILLS

LABORATORY

Mammalian cell culture · Aseptic technique · Adenoviral overexpression · Primer design · RNA extraction · cDNA synthesis · qPCR & ΔΔCq analysis · PCR · DNA cloning · Transformation · Gel electrophoresis · Electroelution

CODING

C · C++ · Python · TypeScript · JavaScript · HTML & CSS · Tailwind CSS · Git · Docker · Linux & Unix · UX/UI design · Algorithms & data structures

TOOLS & COMMUNICATION

Word · Excel · PowerPoint · Slide & figure design · Scientific documentation

HONOURS

- **42 Heilbronn full scholarship** 2023
- **NSRF scholarship**, one of 40 DUTH students 2020
- **100/100 in Physics**, Panhellenic exams, top 0.1% nationally 2018
- **Harvard University CHS** High School Summer Program 2017 ↗
- **Euroscola**, European Parliament: regional representative 2016 ↗
- **"Thalis" Mathematics Award**, Hellenic Mathematical Society 2015 ↗

LEADERSHIP

- **42 Heilbronn Student Council**, elected member 2023
- **Student Cultural Club PYXIDA**, DUTH: elected special secretary 2019–21
- **High school class president** 2015–17

LANGUAGES

Greek	Native
English	C2 · Michigan ECPE ↗
French	C1 · DALF ↗
German	A2/B1